

Width bounds, finite lex resolutions, and standard higher Tor

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For independent mathematical review. Novelty has not been established. Formal scope and remaining algebraic gaps are stated in Section 9.

Abstract

This version freezes the completed standard higher-Tor chain for actual finite-colength lex ideals I with $x \notin I$ in three variables over an arbitrary field. An explicit finite free resolution of the actual quotient has ranks $1, \alpha + 1 + \ell, \alpha + 2\ell, \ell$ and zero residue differentials. The standard derived-functor comparison and an actual coefficient-field base-change equivalence identify these ranks with finite dimensions of $\mathrm{Tor}_i^A(A/\mathfrak{m}, A/I)$, and prove vanishing in all degrees $i \geq 4$. Under the cumulative width budget, all positive-degree binomial bounds and the strict first bound are therefore Lean checked for this lex model. The earlier attained extremum $\Theta(w \log w)$ is preserved. The original numerical-semigroup application remains a conventional proof: its reduction and comparison chain has not been formalized end to end. No general formal interchange of Tor arguments, novelty certification, or human peer-review endorsement is asserted. Versions 0.1–0.3 are preserved.

1 The semigroup statement: conventional proof

A numerical semigroup is an additive submonoid $\Gamma \subseteq \mathbb{N}$ with finite complement, where $\mathbb{N} = \{0, 1, \dots\}$. Suppose Γ has exactly four minimal generators $g_0 < g_1 < g_2 < g_3$, and put $m = g_0$ and $w = g_3 - g_0$. For a field $\mathbb{k}$, write

$$R_\Gamma = \mathbb{k}[[t^{g_0}, t^{g_1}, t^{g_2}, t^{g_3}]].$$

All Betti numbers of R_Γ are over its minimal regular presentation $\mathbb{k}[[X_0, X_1, X_2, X_3]]$.

Theorem 1.1 *For every such Γ and every field $\mathbb{k}$,*

$$\beta_i(R_\Gamma) \leq i \binom{w+1}{i+1} \quad (i \geq 1).$$

If $w \geq 4$, then in addition

$$\beta_1(R_\Gamma) \leq \binom{w+1}{2} - 1.$$

Caviglia–Moscariello–Sammartano (CMS) prove the non-strict bounds for $w \geq 40$ in [1, Theorem 1.5]; their Remark 5.2 discusses the remaining small-width lex problem. Draft v0.1 treated that interval using a finite envelope certificate. The analytic proof works uniformly for all $w \geq 4$ and makes the first bound strict. The case $w = 3$ remains the known consecutive-generator case. We do not claim these stronger conclusions for arbitrary tangent cones or for arbitrary embedding dimension.

2 A terminal budget for lex columns

Let $S = \mathbb{k}[x, y, z]$, ordered lexicographically with $x > y > z$, and let $L \subseteq (x, y, z)^2$ be an artinian lex ideal. Its standard monomials are those outside L . Assume that, for an integer $w \geq 4$,

$$\sum_{t=0}^d \text{HF}(S/L, t) \leq 1 + dw \quad (d \in \mathbb{N}). \quad (1)$$

Put $K = L \cap \mathbb{k}[x, y]$, and let α be the least exponent such that $x^\alpha \in L$. Then $\alpha \geq 2$. For $0 \leq i < \alpha$, define the positive column length

$$n_i = \#\{j \geq 0 : x^i y^j \notin K\}, \quad \ell = \text{length}(\mathbb{k}[x, y]/K) = \sum_{i=0}^{\alpha-1} n_i.$$

Each column is an initial interval in its y exponent. Since $x^i y^{n_i-1}$ is standard, lex order makes y^{i+n_i-1} standard. Hence

$$i + n_i \leq n_0 \quad (0 \leq i < \alpha). \quad (2)$$

Lemma 2.1 *The column lengths satisfy*

$$\sum_{i=0}^{\alpha-1} \frac{n_i(n_i + 1)}{2} \leq 1 + (n_0 - 1)w. \quad (3)$$

Proof. For fixed i , all monomials $x^i y^j z^k$ with $j + k < n_i$ are standard: first $x^i y^{j+k}$ is standard, and replacing some y factors by z lowers lex order. They have total degree at most $i + n_i - 1 \leq n_0 - 1$, by (2). Their number is $n_i(n_i + 1)/2$, and the sets for distinct i are disjoint. Apply (1) once, at $d = n_0 - 1$. $\square$

Lemma 2.2 *One has $\alpha \leq w - 2$.*

Proof. All monomials of degree below α are standard, since the largest such monomials x^d are standard. Thus

$$\binom{\alpha + 2}{3} \leq 1 + (\alpha - 1)w.$$

Using

$$\alpha(\alpha + 1)(\alpha + 2) - 6 = (\alpha - 1)(\alpha^2 + 4\alpha + 6)$$

and $\alpha \geq 2$ gives $\alpha^2 + 4\alpha + 6 \leq 6w$. If $\alpha \geq w - 1$, this implies $w^2 + 2w + 3 \leq 6w$, or $(w - 1)(w - 3) \leq 0$, impossible for $w \geq 4$. The integer parameters therefore satisfy $\alpha \leq w - 2$. $\square$

3 A uniform strict estimate

Lemma 3.1 *Under (3),*

$$(2\ell + \alpha - 2w)^2 \leq \alpha[4(w - 1)(w - 2) + \alpha]. \quad (4)$$

Proof. Use the signed integer vector

$$v_0 = 2n_0 + 1 - 2w, \quad v_i = 2n_i + 1 \quad (1 \leq i < \alpha).$$

There is no nonnegativity assumption on v_0 . Direct expansion and (3) yield

$$\begin{aligned}\sum_i v_i &= 2\ell + \alpha - 2w, \\ \sum_i v_i^2 &= 4 \sum_i n_i(n_i + 1) + \alpha - 8wn_0 - 4w + 4w^2 \\ &\leq 4(w-1)(w-2) + \alpha.\end{aligned}$$

Now apply $(\sum_i v_i)^2 \leq \alpha \sum_i v_i^2$. Equivalently, this step follows from the integer identity

$$\alpha \sum_i v_i^2 - (\sum_i v_i)^2 = \sum_{i < j} (v_i - v_j)^2 \geq 0.$$

□

Theorem 3.2 *For every integer $w \geq 4$ under the hypotheses of Section 2,*

$$\alpha + 1 + \ell \leq \binom{w+1}{2} - 1. \quad (5)$$

Moreover,

$$\alpha + 2\ell \leq 2\binom{w+1}{3}, \quad \ell \leq 3\binom{w+1}{4}. \quad (6)$$

Proof. Set $T = w^2 - w - \alpha - 2$. Lemma 2.2 gives $T \geq w(w-2) > 0$. A polynomial identity and the same lemma show

$$\begin{aligned}T^2 - \alpha[4(w-1)(w-2) + \alpha] &= (w-2)[(w-2)(w+1)^2 - 2\alpha(3w-1)] \\ &\geq (w-2)^2(w-1)(w-3) > 0.\end{aligned}$$

Combining this with (4) gives $2\ell + \alpha - 2w < T$, since $T > 0$. Rearranging yields

$$\alpha + 1 + \ell < \frac{w(w+1)}{2},$$

which is (5) by integrality.

Write $C = \binom{w+1}{2}$. Then $\ell \leq C - \alpha - 2$, so $\alpha + 2\ell \leq 2C - \alpha - 4 \leq 2C$ and $\ell \leq C$. Finally,

$$C \leq \binom{w+1}{3}, \quad C \leq 3\binom{w+1}{4} \quad (w \geq 4),$$

as is immediate from the ratios $(w-1)/3$ and $(w-1)(w-2)/4$. This proves (6) without a bounded parameter search. □

Remark 3.3 *The retained estimate (4) also gives*

$$\ell \leq w - \frac{\alpha}{2} + \sqrt{\alpha(w-1)(w-2) + \frac{\alpha^2}{4}}.$$

Since $\alpha^2 + 4\alpha + 6 \leq 6w$, this also yields a conventional $O(w^{5/4})$ estimate. Section 7 gives the stronger, formally verified logarithmic estimate; the square-root display is retained to explain the intermediate route.

4 The semigroup transfer: conventional layer

For the finite-colength lex ideal L , the two-variable lex ideal K has $\alpha + 1$ minimal generators and rank one, so its Betti numbers as an ideal are $(\alpha + 1, \alpha)$. The finite-colength formula [1, equation (7)] gives

$$\beta_0^S(L) = \alpha + 1 + \ell, \quad \beta_1^S(L) = \alpha + 2\ell, \quad \beta_2^S(L) = \ell. \quad (7)$$

Thus Theorem 3.2 bounds the three positive Betti numbers of S/L , with the usual ideal-to-quotient shift by one. Section 6 now gives an independent, Lean-checked realization of the quotient formulas in the fixed standard-Tor argument order. The semigroup comparisons below remain in the conventional layer.

Proof of Theorem 1.1. The four-generator hypothesis gives $w \geq 3$. If $w = 3$, the generators are consecutive. Proposition 2.7 of [2], with $r = 4$, gives the desired bound for the graded semigroup ring. Localizing its minimal weighted-graded free resolution at the homogeneous maximal ideal and then completing preserve exactness and minimality, hence preserve the Betti numbers.

Suppose now $w \geq 4$. CMS's construction starts with the Artinian reduction $R_\Gamma/(t^m)$ and passes to its associated graded ring and then to a homogeneous monomial initial ideal $J \subseteq Q = \mathbb{k}[x, y, z]$. It has colength m and no linear forms. The regular parameter t^m , the initial-ideal comparison and [1, equation (1)] give

$$\beta_i(R_\Gamma) \leq \beta_i^Q(Q/J) \quad (i \geq 1).$$

If $w \leq m - 2$, Theorem 3.3 of [1] gives $\text{HS}(Q/J, d) \leq 1 + dw$. If $w \geq m - 1$, the same inequality follows directly: the degree-zero contribution is one, and for $d \geq 1$,

$$\text{HS}(Q/J, d) \leq m \leq w + 1 \leq 1 + dw.$$

Thus the budget holds for every $w \geq 4$, without invoking a separate large-width theorem.

Let $L = \text{Lex}(J)$. It remains artinian, has no linear forms and has the same Hilbert function. The characteristic-independent Bigatti–Hulett–Pardue comparison [1, Theorem 2.1] implies

$$\beta_i(R_\Gamma) \leq \beta_i^Q(Q/J) \leq \beta_i^Q(Q/L).$$

Apply (7) and Theorem 3.2 for $i = 1, 2, 3$. For $i \geq 4$, the Betti numbers vanish: R_Γ is one-dimensional Cohen–Macaulay with a four-dimensional minimal regular presentation, and therefore has projective dimension three. This proves both conclusions. $\square$

The Artinian assumption in (7) is not inferred from the combinatorial theorem. It comes from the semigroup reduction. For example, (x^2, xy, xz, y^2, yz) retains infinitely many standard powers of z and does not satisfy the finite-colength Betti formula. The general lex statement at $w = 3$ is also false; that boundary is handled separately for semigroups.

5 Actual ideals, generator numbers, and standard Tor

The following results describe the present machine-checked algebraic layer. Write $A = \mathbb{k}[x, y, z]$ and $\mathfrak{m} = (x, y, z)$, where $\mathbb{k}$ is any field. A monomial ideal is represented by an exponent upper set $E \subseteq \mathbb{N}^3$; membership of a monomial in the ideal is proved equivalent to membership of its exponent in E . Lex closure refers to each fixed total degree, with $x > y > z$. The notation ℓ now also denotes the dimension of the actual image of $\mathbb{k}[x, y]$ in A/I , rather than a count awaiting interpretation.

Theorem 5.1 (Actual quotient dimensions and generator number) *Let I be a finite-colength lex monomial ideal, with $x \notin I$, satisfying (1) for $w \geq 4$. Let α be the least pure- x exponent in I . Then the standard monomial classes form a basis of A/I ; the dimension of each homogeneous image is its standard-monomial count. The full xy image has $\mathbb{k}$ -dimension $\ell = \sum_{i < \alpha} n_i$. There is an explicit complete set of divisibility-minimal monomial generators of I , of cardinality $\alpha + 1 + \ell$. For every finite set $P \subset A$ with $(P) = I$, including nonhomogeneous polynomial sets,*

$$|P| \geq \alpha + 1 + \ell.$$

Equality is attained by that monomial set. Thus the actual least generator number satisfies $\mu_A(I) = \alpha + 1 + \ell$, and obeys (5).

The quotient bases come from explicit coefficient maps with the actual monomial ideal as kernel. Finiteness of the quotient is equivalent to finiteness of the set of standard exponents; for lex ideals it is also equivalent to the presence of a pure power of z . The lower bound for arbitrary generating sets uses the coefficients at all minimal exponents. Multiplication on those coordinates acts through the constant coefficient. A generating family therefore spans the whole coordinate space, forcing the cardinality bound. This proves a least cardinality, rather than merely deletion-irredundance.

Theorem 5.2 (Genuine fiber and standard first Tor) *For any ideal $I \subseteq A$ there is a $\mathbb{k}$ -linear isomorphism*

$$(A/\mathfrak{m}) \otimes_A I \simeq I/\mathfrak{m}I,$$

and $A/\mathfrak{m} \simeq \mathbb{k}$ as $\mathbb{k}$ -algebras. If $I \subseteq \mathfrak{m}$, then

$$\mathrm{Tor}_1^A(A/\mathfrak{m}, A/I) \simeq (A/\mathfrak{m}) \otimes_A I \simeq I/\mathfrak{m}I.$$

For I as in Theorem 5.1, these spaces have dimension $\alpha + 1 + \ell = \mu_A(I)$. Their bases correspond to the minimal monomial classes and the pure tensors $1 \otimes g$.

The first Tor here is mathlib's standard categorical Tor: the first argument $A/\mathfrak{m}$ is fixed and the functor is derived in the second argument. No interchange of the two arguments is being assumed as a formal theorem. The construction uses the actual short exact sequence $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$, a projective resolution starting with the specified surjection, and the standard left-derived comparison. It does not assume that I is projective. Restriction to $\mathbb{k}$ uses the original algebra map and its scalar action, not an action defined by transport across the desired isomorphism.

The condition $I \subseteq \mathfrak{m}$ in the Tor statement is essential: $I = A$ has nonzero fiber $A/\mathfrak{m}$ but $\mathrm{Tor}_1^A(A/\mathfrak{m}, 0) = 0$. Likewise, the equality between fiber dimension and the global minimum number of generators is asserted here with the complete minimal monomial hypotheses; it is not a theorem about arbitrary polynomial ideals.

6 A finite lex resolution and standard higher Tor

This section is the new formal milestone. It uses the actual quotient A/I and the standard Tor functor throughout. Let I satisfy the hypotheses of Theorem 5.1; write n_i for its complete xy column lengths. For $0 \leq i < \alpha$ and $0 \leq j < n_i$, let t_{ij} be the least nonnegative integer for which $x^i y^j z^{t_{ij}} \in I$. Such an integer exists by finite colength, and it is positive because $x^i y^j$ is standard.

Theorem 6.1 (Actual finite free resolution) *There is an actual standard projective resolution of A/I with finite free terms, represented by the exact augmented sequence*

$$0 \longrightarrow A^\ell \xrightarrow{d_3} A^{\alpha+2\ell} \xrightarrow{d_2} A^{\alpha+1+\ell} \xrightarrow{d_1} A \xrightarrow{q} A/I \longrightarrow 0.$$

Here q is the original quotient map. All entries of d_1, d_2, d_3 belong to $\mathfrak{m}$, and the terms in degrees at least four are actual zero objects. The augmentation is a proved quasi-isomorphism; the displayed ranks do not serve as assumptions defining a resolution.

Construction and exactness. The first free basis is the complete minimal boundary family

$$x^\alpha, \quad x^i y^{n_i} \ (i < \alpha), \quad x^i y^j z^{t_{ij}} \ (i < \alpha, \ j < n_i).$$

It has $\alpha + 1 + \ell$ elements. The map d_1 sends these basis vectors to the corresponding actual monomials; their span equals I , so $\text{im } d_1 = \ker q$.

The second basis has one x -step for each xy boundary and two steps, in the x and y directions, for each z boundary. Its size is $\alpha + 2\ell$. For a step from exponent u in direction v , its common degree is $\delta = u + e_v$. A specified canonical boundary exponent $c(\delta) \leq \delta$ gives the column below, where x_v is the step variable and $\mathbf{x}^\nu$ denotes the three-variable monomial:

$$\rho_{u,v} = x_v e_u - \mathbf{x}^{\delta - c(\delta)} e_{c(\delta)}.$$

This is an actual kernel vector of d_1 . The proof of generation handles arbitrary polynomial vectors in the kernel: common-degree monomial differences generate all relations, and canonical boundary steps normalize each difference by a strictly decreasing finite height. This proves the full equality $\text{im } d_2 = \ker d_1$, not just $d_1 d_2 = 0$. The quotient argument used to prove generation requires only a coefficient-field linear lift; it does not assume an A -linear splitting of $A \rightarrow A/I$.

The third basis is indexed by the ℓ standard xy positions. At a z boundary p , the corresponding x - and y -steps have indices $r_x(p), r_y(p)$. The actual column has the form

$$f_p = y e_{r_x(p)} - x e_{r_y(p)} - (c_y - c_x).$$

The correction vectors are constructed by the preceding normalization, not supplied as unproved data. They have strictly lower support height, all their coefficients lie in $\mathfrak{m}$, and $d_2(c_y - c_x) = d_2(y e_{r_x(p)} - x e_{r_y(p)})$. Hence $d_2 f_p = 0$, with leading entries $y, -x$.

For a second syzygy, its highest remaining row gives $xf + yg = 0$. Polynomial coefficient divisibility and domain cancellation prove $f = yh, g = -xh$. Subtracting hf_p lowers the height, and finite elimination gives $\text{im } d_3 = \ker d_2$. The same triangular leading entries prove d_3 injective. No claim that the ideal (x, y) is the unit ideal is used. Degree control in the step and correction constructions puts every differential entry in $\mathfrak{m}$. The concrete exactness and injectivity data then produce mathlib's standard projective resolution with the original quotient augmentation. $\square$

Theorem 6.2 (Standard Tor dimensions and vanishing) *Define*

$$T_i(I) = \text{Tor}_i^A(A/\mathfrak{m}, A/I), \quad b_i(I) = \dim_{\mathbb{k}} T_i(I),$$

where $T_i(I)$ has scalars restricted along the actual inclusion $\mathbb{k} \rightarrow A$. Every $T_i(I)$ is finite dimensional over $\mathbb{k}$, and

$$(b_0, b_1, b_2, b_3) = (1, \alpha + 1 + \ell, \alpha + 2\ell, \ell), \quad T_i(I) = 0 \quad (i \geq 4).$$

Consequently, for every $w \geq 4$ satisfying the actual cumulative budget,

$$b_i(I) \leq i \binom{w+1}{i+1} \quad (i \geq 1), \quad b_1(I) \leq \binom{w+1}{2} - 1.$$

The zero assertion is a standard categorical zero object, stronger than a bare computation of a natural-valued rank.

Proof. For the actual resolution P , mathlib's standard left-derived comparison identifies $T_i(I)$ with $H_i((A/\mathfrak{m}) \otimes_A P)$, deriving the second argument A/I . All tensor differentials vanish by Theorem 6.1. The zero-differential homology data therefore give an A -module isomorphism from this actual homology object to $(A/\mathfrak{m}) \otimes_A P_i$.

For any finite free A -module F , a genuine finite free basis gives

$$(A/\mathfrak{m}) \otimes_A F \simeq_{\mathbb{k}} (A/\mathfrak{m})^B \simeq_{\mathbb{k}} \mathbb{k}^B.$$

The first coordinate map is first A -linear and then restricted to $\mathbb{k}$; the second uses the proved residue-algebra isomorphism. This independently proves $\mathbb{k}$ -finiteness and then identifies the $\mathbb{k}$ -dimension with the A -rank of F . The free term F itself is not asserted to be finite dimensional over $\mathbb{k}$. Applying this to each P_i proves the four formulas. Zero terms and preservation of zero objects give $T_i(I) = 0$ for $i \geq 4$.

The initial exponent from the resolution equals that in the column inequalities because both are the first forbidden pure- x exponent. Theorem 3.2 thus gives the three positive-degree inequalities; the zero tail gives every remaining degree. At degree one, the standard object and coefficient action agree definitionally with Theorem 5.2; its earlier generator-fiber dimension formula remains valid for the same object. $\square$

The construction also has a budget-free interface: actual finite boundary data, lex closure, the proved monomial span equality, and $I \subseteq \mathfrak{m}$ give ranks $1, \alpha + 1 + \sum n_i, \alpha + 2 \sum n_i, \sum n_i$ and the same Tor calculation without a width parameter. The original finite-colength budget model supplies all those data. The precise minimality verified here means that every residual differential is zero; a separate full graded-shift API is not claimed. No general formal exchange of the two Tor arguments is used or supplied.

7 Harmonic growth and a realized lower family

Let $H_n = \sum_{r=1}^n 1/r$, with $H_0 = 0$. All inequalities involving H_n were first proved over the rationals, with exact natural-number counts. For every ideal under Theorem 5.1, the formal development gives

$$\ell \leq 3w + (2w - 1)H_{2w-1}, \quad \mu_A(I) \leq 4w - 1 + (2w - 1)H_{2w-1}. \quad (8)$$

The second inequality uses $\alpha \leq w - 2$; it also bounds the dimension of the standard Tor space in Theorem 5.2. Comparison with the library harmonic numbers and the natural logarithm yields

$$\ell \leq 3w + (2w - 1)(1 + \log(2w - 1)) \leq 10w \log w \quad (w \geq 4). \quad (9)$$

The constant 10 is explicit and convenient; no optimality is claimed.

For $s \geq 2$, define the following actual monomial ideal:

$$I_s = (x^i y^j z^k : (i + 1)(i + j + k + 1) > s^2).$$

The exponent condition is upward closed and lex in each degree. The quotient is finite dimensional: all monomials of degree at least s^2 belong to I_s . The least pure- x exponent is s . Its degree- d xy count is

$$h_d = \min \left(d + 1, \left\lfloor \frac{s^2}{d + 1} \right\rfloor \right).$$

For the three-variable degree count $H_d^{(s)}$, the injection $(i, j) \mapsto i(d + 1) + j$ on standard exponent triples $(i, j, d - i - j)$ proves $H_d^{(s)} \leq s^2$; also $H_0^{(s)} = 1$. Consequently

$$\sum_{t=0}^d H_t^{(s)} \leq 1 + ds^2 \leq 1 + dw \quad \text{whenever } s^2 \leq w.$$

These are dimensions of the actual quotient homogeneous images by Theorem 5.1's interface, not a proposed Hilbert sequence without an ideal realizing it.

Theorem 7.1 (Exact lower-family invariants) *The actual xy image for I_s has dimension*

$$\ell_s = \sum_{i=0}^{s-1} \left(\left\lfloor \frac{s^2}{i+1} \right\rfloor - i \right), \quad s^2 H_s \leq \ell_s + \frac{s^2 + s}{2}.$$

Its least number of polynomial generators and its standard first Tor dimension are both $s + 1 + \ell_s$.

Each summand is nonnegative; the implementation accounts for truncated natural subtraction explicitly. This family supplies a lower bound in the abstract lex budget class. No realization of these ideals by numerical semigroups is claimed.

8 An attained extremum and standard Theta

For each natural number w , let $\mathcal{C}_{\mathbb{k}, w}$ consist of actual monomial ideals I of $A = \mathbb{k}[x, y, z]$ such that:

1. the defining exponent set is upward closed and lex;
2. A/I is finite dimensional over $\mathbb{k}$ and $x \notin I$;
3. every degree satisfies the cumulative actual-dimension budget $\sum_{t=0}^d \dim_{\mathbb{k}}(A/I)_t \leq 1 + dw$.

Define $M_{\mathbb{k}}(w)$ as the natural supremum of the dimensions of the actual xy images for ideals in this class. For $w \geq 4$, I_2 makes the class nonempty, and the strict binomial estimate bounds its set of dimensions. A nonempty bounded set of natural numbers has a greatest member. Thus this supremum is attained by an actual ideal, rather than an independent sum of optimal per-degree envelopes.

Theorem 8.1 (Formal extremal growth) *For any field $\mathbb{k}$, the maximum $M_{\mathbb{k}}(w)$ is attained for $w \geq 4$, and*

$$\frac{w \log w}{16} \leq M_{\mathbb{k}}(w) \leq 10w \log w \quad (w \geq 64).$$

As a function on natural widths with values in $\mathbb{R}$, $M_{\mathbb{k}}(w) = \Theta(w \log w)$ at infinity, in the standard mathlib `Asymptotics.IsTheta` sense.

Proof structure. The upper estimate is (9) applied to a maximizing ideal. For each $w \geq 64$, choose $s = \lfloor \sqrt{w} \rfloor$. Then $s \geq 2$ and $s^2 \leq w$, so $I_s \in \mathcal{C}_{\mathbb{k}, w}$. The exact floor-sum estimate in Theorem 7.1, harmonic-log comparison, and integer square-root bounds give $\ell_s \geq w \log w / 16$. All rounding and scalar conversions are checked in Lean. This proves the lower estimate for every sufficiently large integer w , not just squares. The two inequalities give the two standard Big-O relations, including the required norms and eventual threshold. $\square$

The real logarithm is used only for numerical bounds on dimensions. The coefficient field need not have characteristic zero. The theorem describes an extremum over the stated class; it is not a growth lower bound for each member, nor for numerical semigroup Betti numbers. The upper comparison from semigroups to lex ideals cannot be reversed to obtain such a lower bound.

9 Formal verification, evidence and limits

This version closes the actual three-variable lex higher-Tor chain of Section 6. It does not claim a full formalization of Theorem 1.1. The conventional semigroup proof in Section 4 still uses parameter reduction, associated-graded and initial-ideal comparisons, and the characteristic-independent lex comparison from the cited literature. Those bridges, the semigroup $w = 3$ branch, and any necessary comparison of Tor argument conventions remain substantive formal gaps. The lex quotient formulas and its high-degree vanishing are now checked; the original four-variable complete-local semigroup theorem is not thereby formalized.

Result	Status at delivery
Uniform strict combinatorial bounds	Lean checked; no finite-certificate dependency
Actual monomial quotient dimensions	Lean checked
Least arbitrary polynomial generator number	Lean checked, stated monomial class
Actual fiber, residue field, standard Tor_1	Lean checked, with $I \subseteq \mathfrak{m}$ for Tor
Actual finite free lex quotient resolution	Lean checked; exactness, zero tail, residue minimality
Standard higher Tor dimensions and all positive bounds	Lean checked in the actual lex budget model
Explicit lower family and exact floor sum	Lean checked
Attained maximum and standard $\Theta(w \log w)$	Lean checked, abstract lex budget class
Semigroup width theorem and higher Betti transfer	Conventional proof, internally reviewed
Finite examples (267 parameter pairs; 1807 semigroups)	Diagnostic experiments, not a general proof
Novelty and external expert review	Unestablished; external review deferred

Source entry points. The project entry is `lean/WidthBounds.lean`. The main modules include `AllWidths`, `TorOneBounds`, `LexGrowthTheta`, `LexQuotientResolution`, `ResidueFreeDimension`, `MinimalResolutionTor`, and `HigherTorBounds`; all are under `lean/WidthBounds/`. Exact declarations, supporting modules, hypotheses and evidence are mapped in `research/PROJECT_REPORT.md`. The standard asymptotic endpoint is `maxLexLength_isTheta` in namespace `WidthBounds.LexExtremal`. The declaration source is authoritative.

Verification record. Lean and mathlib are fixed at 4.22.0. The latest successful unified build is `results/lean_higher_tor_build.txt`. Its SHA-256 is

29b05fbcfa9054df43f5b171a511eb51
e28e5745188c4a3a4ce9320fcee787e4.

The two displayed lines concatenate to one digest (the final full stop is not part of it). Accepted source/log hashes are recorded in `research/claims.json`. This delivery changes documentation and packaging only; it rechecks correspondence with the accepted sources instead of claiming a fresh Lean build. To reproduce the formal checks, run `lake build` in `lean/` with the pinned dependency manifest. The delivery guide gives environment and artifact checks separately.

Axiom output for the sixteen new R058 declarations contains only `propext`, `Classical.choice`, and `Quot.sound`. The project disallows proof placeholders, custom axioms and `native_decide`. Actual declaration-dependency audits of seventeen new R058 roots exclude the old finite/small-width certificates and `sorryAx`. These facts describe the checked declarations and their dependencies, not a human endorsement of the intended mathematical model. Independent internal AI reviews inspected definitions and hypotheses; they are not human peer review.

What preservation checks mean. The final packet contains source, task history, logs, claims, this report and historical frozen artifacts. An internal manifest hashes every payload entry; verification checks the archive bytes and recorded evidence references. Hash/ZIP validation does not compile Lean, prove mathematics, authenticate an author, or replace independent reading. Dependency runtimes and private working caches are excluded. The Chinese project report documents the full route, failed assumptions, recovery protocol and bounded future work packages. The delivery guide states exact contents, commands and intentional exclusions.

Closure and optional continuation. The user selected research-delivery closure: no further mathematical task is required for this phase. A later, separately scoped continuation could target actual semigroup reductions or sharper extremal constants. Each needs its own precise hypotheses, tests and new version evidence. No such extension, external contact or publication is initiated by this delivery. A bounded earlier primary-source search did not locate the same conclusion; that fact does not certify novelty. Versions 0.1–0.3 remain historical snapshots of their earlier proof boundaries.

References

- [1] G. Caviglia, A. Moscariello and A. Sammartano, *Bounds for syzygies of monomial curves*, Proc. Amer. Math. Soc. **152** (2024), 3665–3678. doi:10.1090/proc/16862. Version used: arXiv:2307.05770v2, 20 July 2024.
- [2] J. Herzog and D. I. Stamate, *On the defining equations of the tangent cone of a numerical semigroup ring*, J. Algebra **418** (2014), 8–28. doi:10.1016/j.jalgebra.2014.07.008. Version used: arXiv:1308.4644v3, 29 July 2014.